

Chapter-4: Gradient, Divergence and Curl

Gradient

Q-5. Show that $\vec{\nabla}\phi$ is a vector perpendicular to the surface $\phi(x, y, z) = c$ where c is a constant.

Q-5. Show that $\vec{\nabla}\phi$ is a vector normal to the surface $\phi(x, y, z) = c$ where c is a constant.

Q-6. Find a unit normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$.

Exercise- 58, 59

Q-7. Find an equation for the tangent plane to the surface $2xz^2 - 3xy - 4x = 7$ at the point $(1, -1, 2)$.

Exercise- 60, 61

Directional derivative:

Q-10. Find the directional derivative to the surface $\phi = x^2yz + 4xz^2$ at the point $(1, -2, -1)$ in the direction $2\hat{i} - \hat{j} - 2\hat{k}$.

Exercise- 62, 63

Q-12. Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 9$ at the point $(2, -1, 2)$.

Exercise- 65, 66, 67

Write down the geometrical explanation of $\vec{\nabla}\phi$.

Q-11. (a) In what direction from the point $(2, 1, -1)$ is the directional derivative of the surface $\phi = x^2yz^3$ a maximum? What is the magnitude of this maximum?

Exercise- 64

Divergence

Q-21. A fluid moves so that its velocity at any point is $\vec{v}(x, y, z)$. Show that the gain of fluid per unit volume per unit time in a small parallelepiped having centre at $P(x, y, z)$ and edges parallel to the coordinate axes and having magnitude $\Delta x, \Delta y, \Delta z$ respectively, is given approximately by $\text{div } \vec{v}$ or $\vec{\nabla} \cdot \vec{v}$.

Q-22. Determine the constant a so that the vector $\vec{v} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x + az)\hat{k}$.

Exercise- 84, 85

Curl

***Discuss graphically the curl of the force field $\vec{F} = y\hat{i} - x\hat{j}$.

Q-102. Show that $\vec{A} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$ is irrotational. Find ϕ such that $\vec{A} = \vec{\nabla}\phi$.

Exercise- 32, 102, 103

Examples: 25, 26, 27, 29